BBL84

"The L is for Lewis"

Inspiration

Broadband silicon photonic directional coupler using asymmetric-waveguide based phase control

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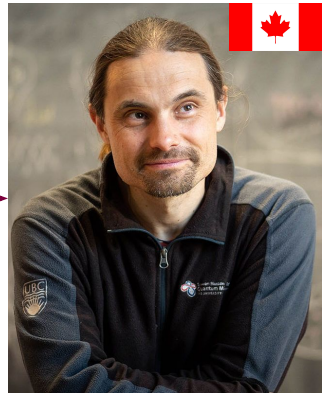
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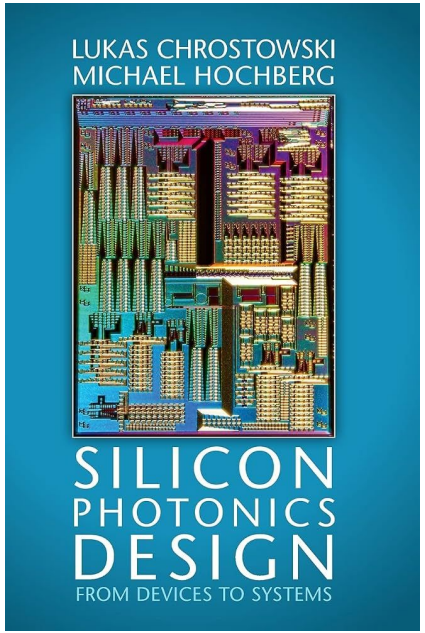
Abstract: We design and demonstrate broadband directional couplers that use asymmetric-waveguide based phase control sections, on the silicon-on-insulator platform. Broadband directional couplers with various power splitting ratios, including 10%/90%, 20%/80%, 30%/70%, 40%/60% and 50%/50%, were realized for both transverse electric (TE) and transverse magnetic (TM) modes. Some of the devices exhibit bandwidths in excess of 100 nm, and all in excess of 75 nm. The footprints of the TE mode couplers are 32 μm $\times$1.3 μm, or less, and those of the TM mode couplers are 13 μm $\times$1.3 μm, or less.

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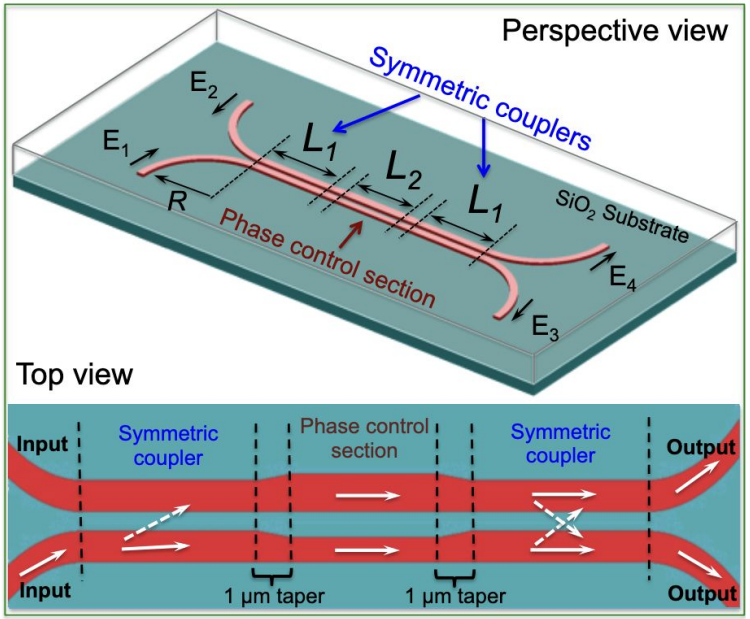
OCIS codes: (130.3120) Integrated optics devices; (230.7370) Waveguides.



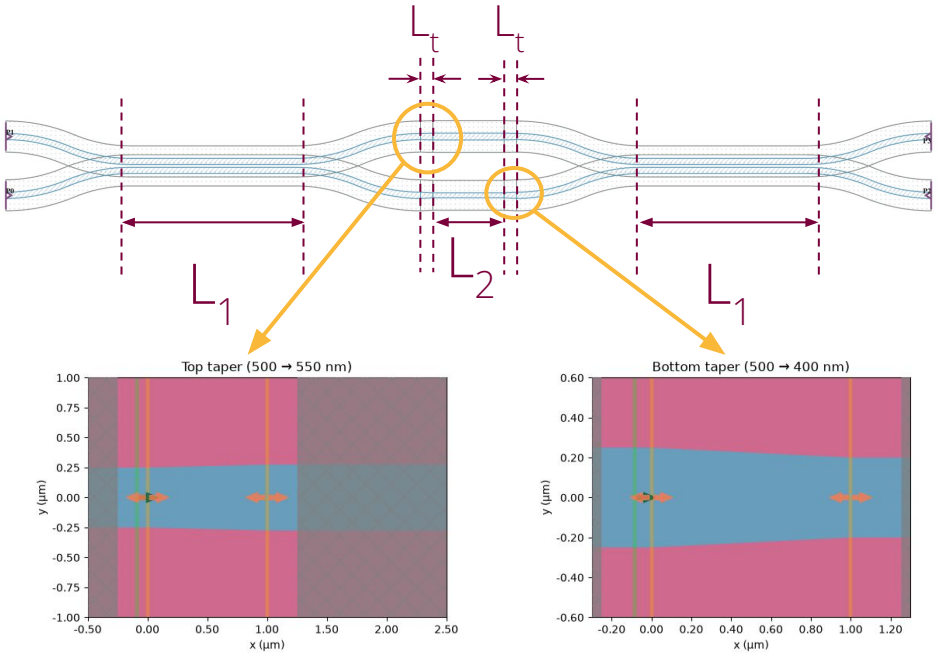
Wrote



Design



"Broadband" Directional-Coupler



Our Implementation

Transfer Matrix Method (TMM)

$$\begin{bmatrix} E_3 \\ E_4 \end{bmatrix} = C \cdot P_t \cdot P \cdot P_t^{-1} \cdot C \cdot \begin{bmatrix} E_1 \\ E_2 \end{bmatrix}$$

Coupling Matrix of Symmetric Couplers

$$C = \begin{bmatrix} t & -jk \\ -jk & t \end{bmatrix} \cdot e^{-j\frac{\pi}{\lambda}(n_+ + n_-)L_1 - \frac{\alpha}{2}\cdot L_1}$$

Phase-Control Propagation Matrix

$$P = \begin{bmatrix} e^{-j\frac{2\pi n_1}{\lambda}\cdot L_2 - \frac{\alpha}{2}\cdot L_2} & 0 \\ 0 & e^{-j\frac{2\pi n_2}{\lambda}\cdot L_2 - \frac{\alpha}{2}\cdot L_2} \end{bmatrix}$$

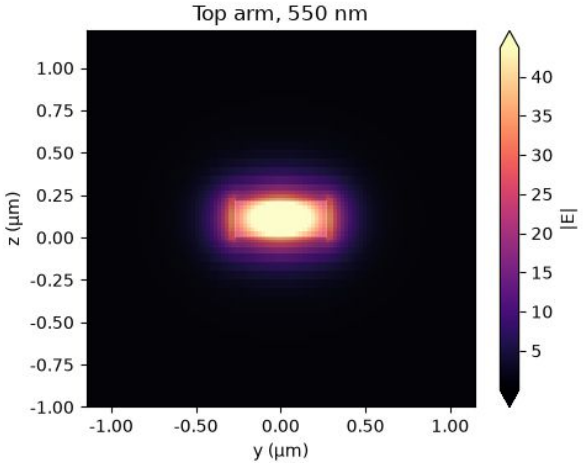
Tapered-Waveguide Propagation Matrix

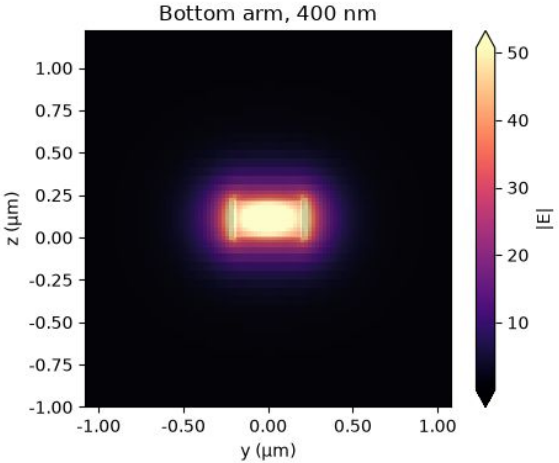
$$P_t = \begin{bmatrix} e^{-j\theta_{t1} - \frac{\alpha}{2}\cdot L_t} & 0 \\ 0 & e^{-j\theta_{t2} - \frac{\alpha}{2}\cdot L_t} \end{bmatrix}$$

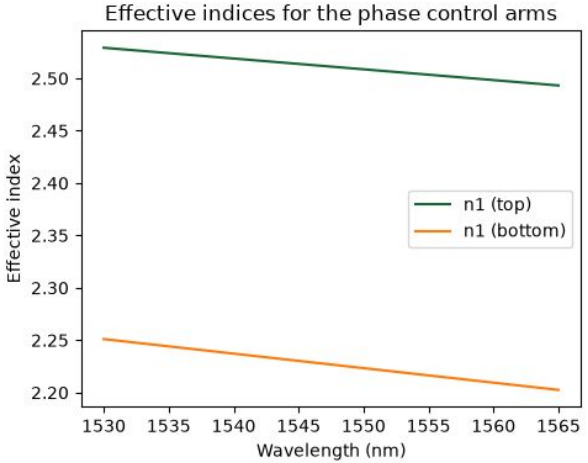
Power Splitting Ratios:

$$\eta_{cross} = \frac{|E_4|^2}{|E_3|^2 + |E_4|^2}\ ; \quad \eta_{through} = \frac{|E_3|^2}{|E_3|^2 + |E_4|^2}$$

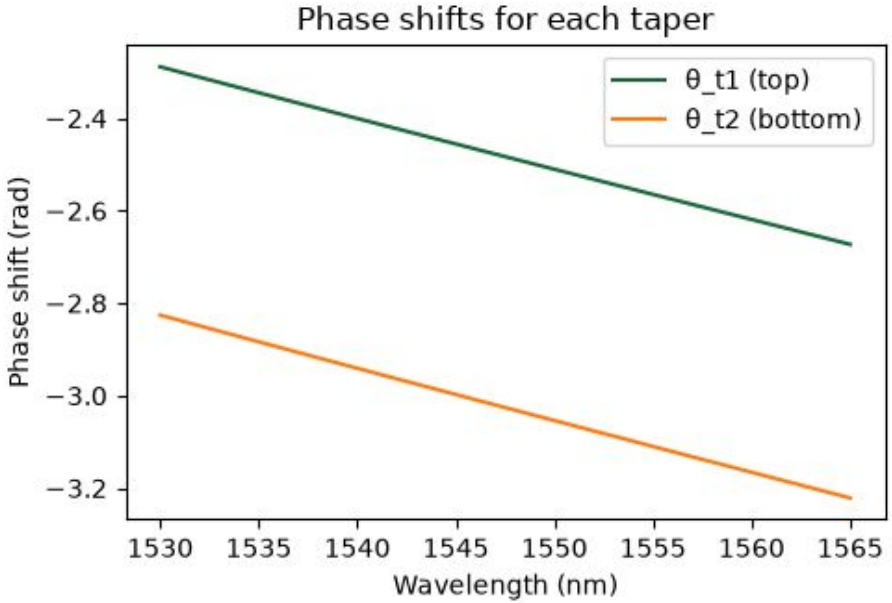
Transfer Matrix Method (TMM) - Results

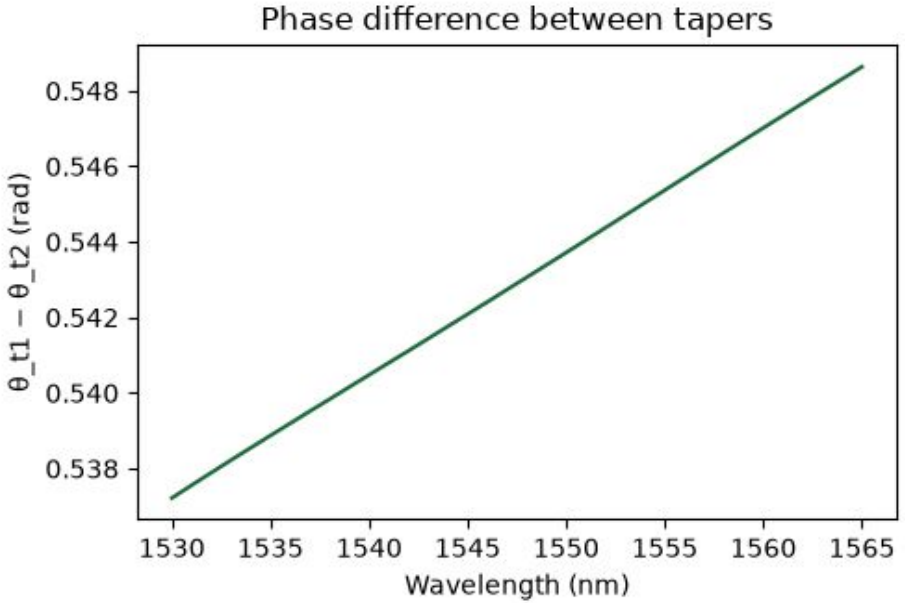






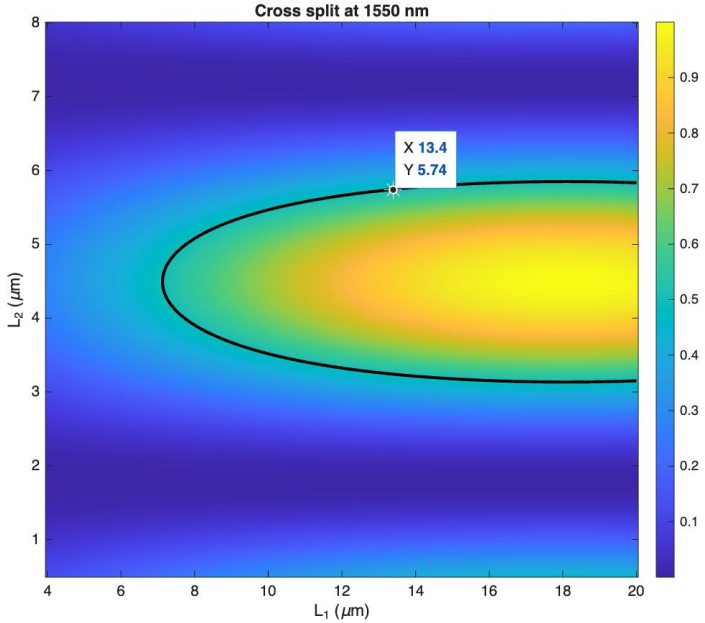
Transfer Matrix Method (TMM) - Results

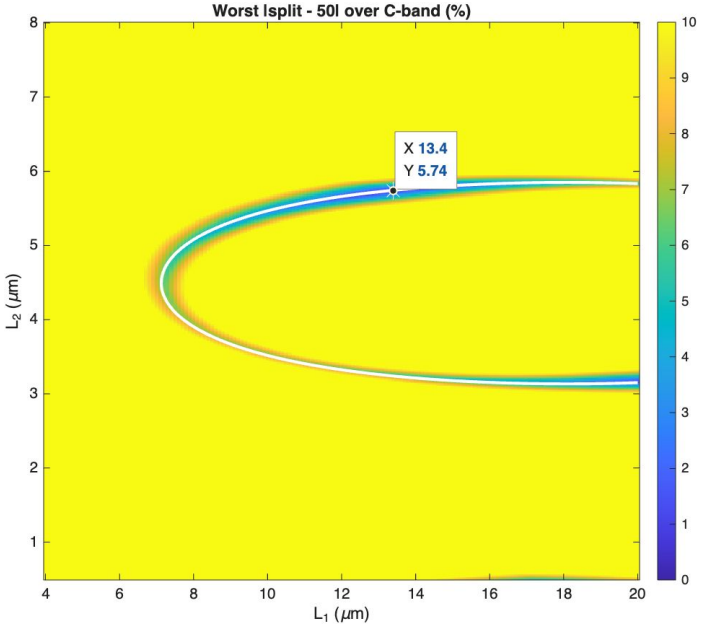




Transfer Matrix Method (TMM)

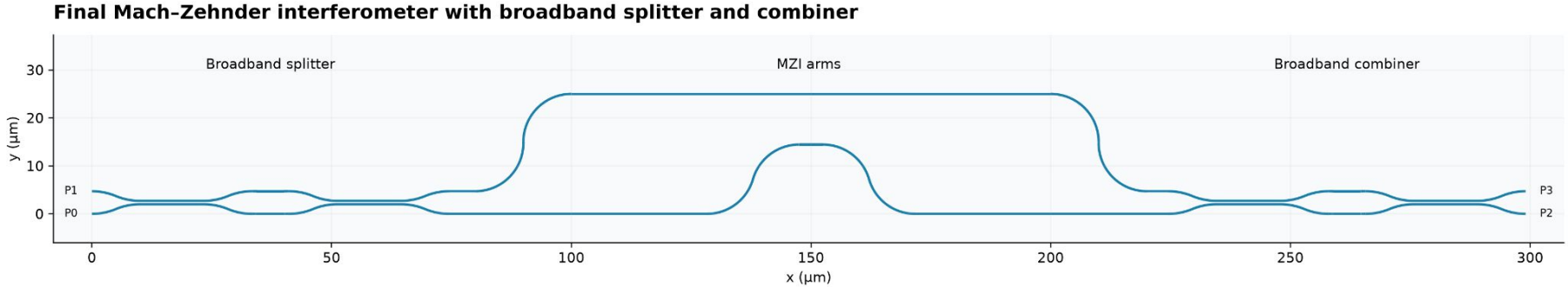
MatLab was used to sweep L_1 and L_2

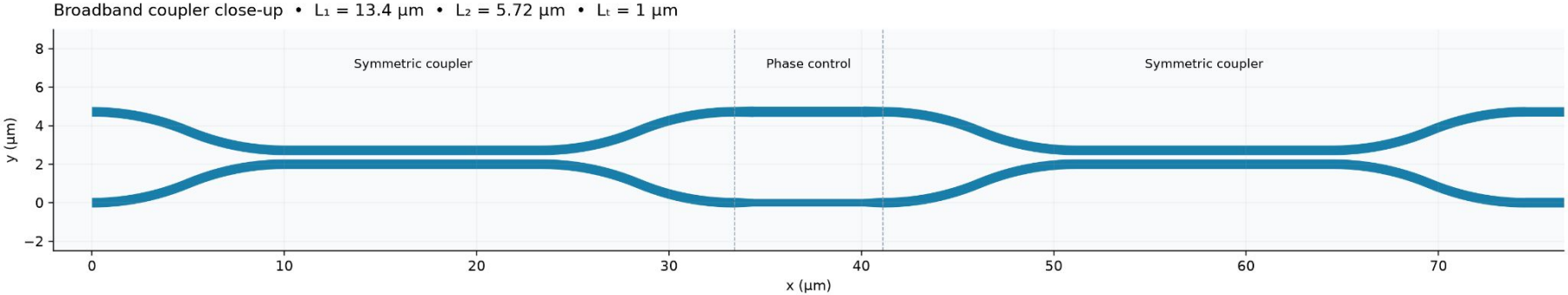




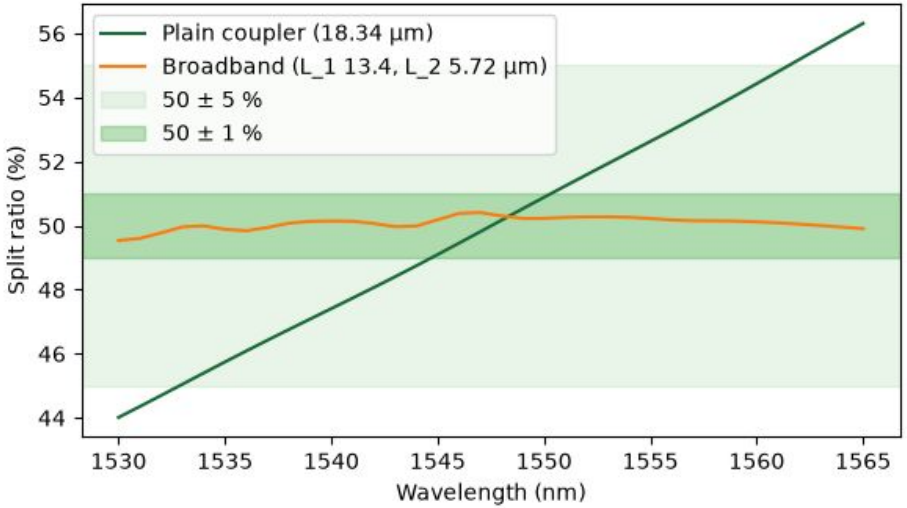
Final MZI Schematic

Distance between couplers: 150 µm | Arm length difference: 10.2422 µm | Phase-arm widths: 550 / 400 nm





Transfer Matrix Method (TMM) - Results



Best ΔL: ***10.2422 µm***

Extinction ratio ***46.8 dB***

PASS Coupler split ratio at 1550 nm within 50 ± 1 %

PASS Minimum gap or feature at least 100 nm

PASS Minimum bend radius at least 5 µm

PASS Stretch: split ratio within 50 ± 5 % from 1530 to 1565 nm

Broadband score: 45.0

What we tried

1. Hand calculations for straight coupler
   → Within 5% for 1550 nm
2. Parameter optimization with scipy [COBYLA]
   a. Minimized the RMS difference between the 50:50 coupling and the current value across the band
   b. Plateaued around 50 ±6%
3. Implemented the paper
   → Optimized with TMM

Next Time

Use MZIs made with our Broadband Directional Coupler to implement photonic quantum circuits!